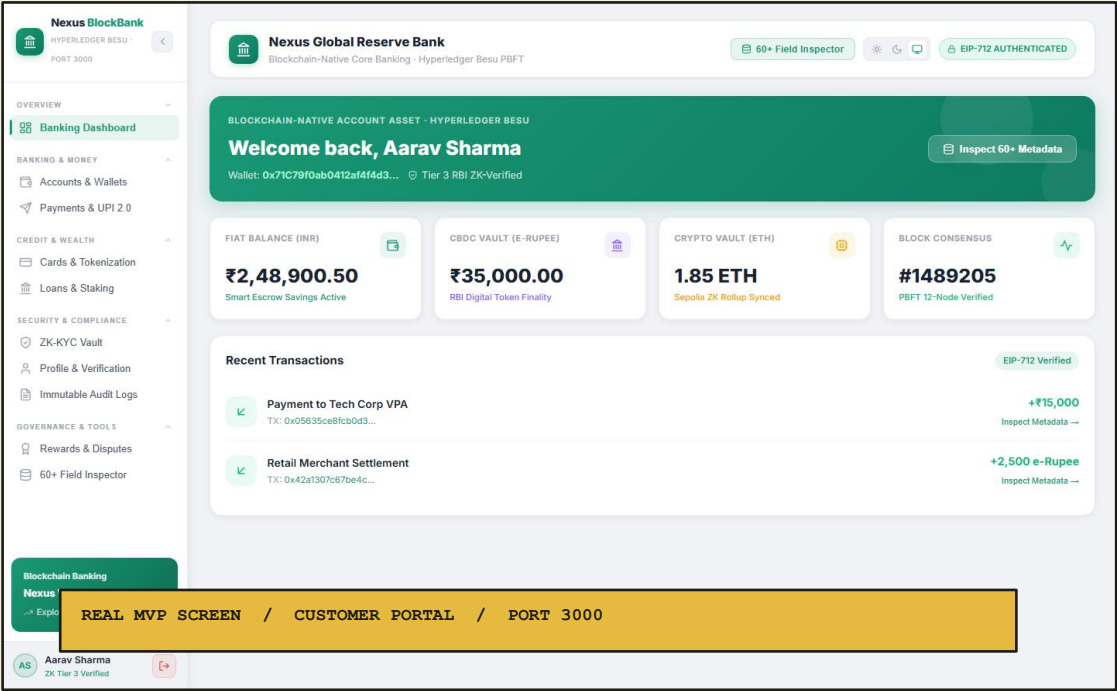


SPECIAL EDITION

Banking that knows the human, not just the password.

Nexus BlockBank adds a continuous trust layer to everyday banking—so sensitive actions are checked, explained and recorded in real time.



<15 ms

REPOSITORY DEMO TARGET FOR TRUST EVALUATION

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AUTOMATED TEST SUITES PASSED LOCALLY

■ INSIDE

Live trust scores. Step-up controls. Dual-portal governance. Proofs for every critical action.

PROBLEM STATEMENT

A valid login can still become a dangerous banking session.

THE RISK BEGINS AFTER LOGIN.

01 LOGIN ≠ TRUST

Passwords and one-time checks cannot judge every transfer, device change or high-risk action.

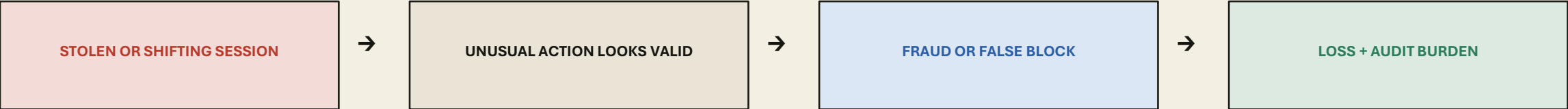
02 SILOED CONTROLS

Customer banking, fraud review and regulator oversight often live in separate tools.

03 WEAK EVIDENCE

A blocked action without clear reasons creates disputes, slow reviews and audit friction.

IMPACT CHAIN

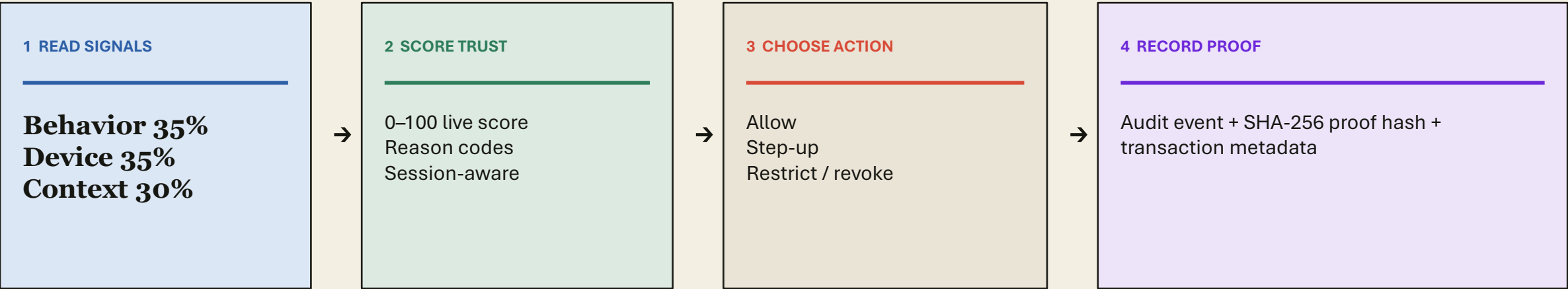


THE REAL QUESTION: DOES THIS ACTION STILL LOOK SAFE—RIGHT NOW?

SOLUTION

One trust decision for every sensitive action.

Nexus BlockBank quietly reads context, scores risk, takes the least disruptive action and leaves an explainable proof.



THE DIFFERENTIATOR

LOW FRICTION FOR GENUINE USERS

Authentication is no longer an entry gate. It becomes a continuous, evidence-based conversation with every banking action.

WORKING PROTOTYPE

The trust layer stays quiet— until risk appears.

A familiar banking dashboard covers money movement, identity and proof without forcing users into a separate security tool.

EVERYDAY BANKING

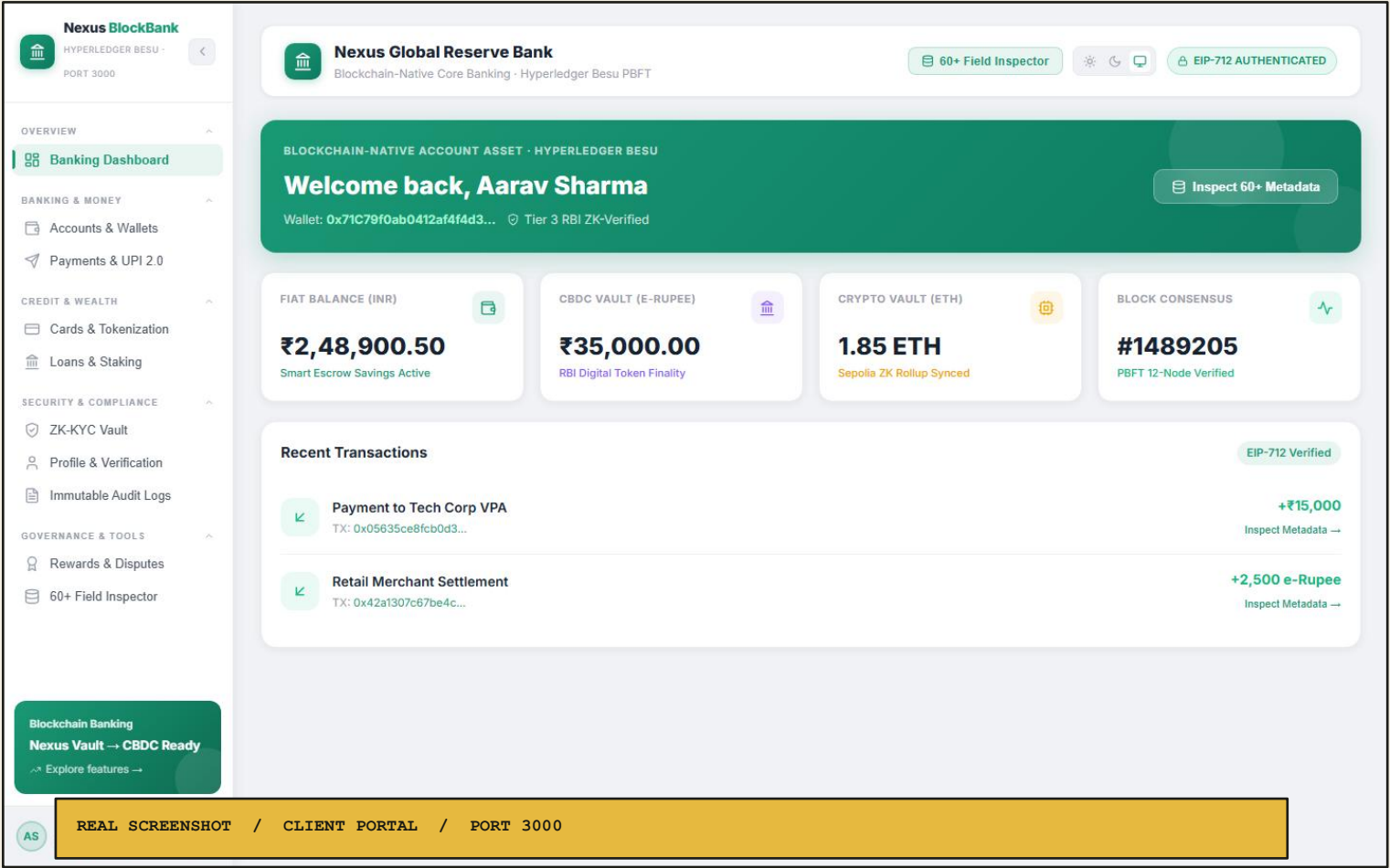
Accounts · UPI · cards · loans

IDENTITY

ZK-KYC vault · profile verification

TRANSPARENCY

Trust badge · immutable audit logs



WHEN RISK RISES: STEP-UP VERIFICATION APPEARS. WHEN TRUST RETURNS: BANKING CONTINUES.

TRUST ENGINE

Three signals. Four safe outcomes.

The rules are intentionally simple enough to explain to a customer, investigator or regulator.

80–100	ALLOW	Low risk	Known device + strong signals
50–79.9	STEP UP	Medium risk	Ask for extra proof
0–49.9	RESTRICT	High risk	Pause the sensitive action
REVOKED	REVOKE	Admin override	Freeze access immediately

WHAT THE ENGINE EXPLAINS

01 KNOWN ATTESTED DEVICE

02 UNUSUAL IP OR TIME

03 ANOMALOUS BEHAVIOR

04 SESSION TAKEOVER RISK

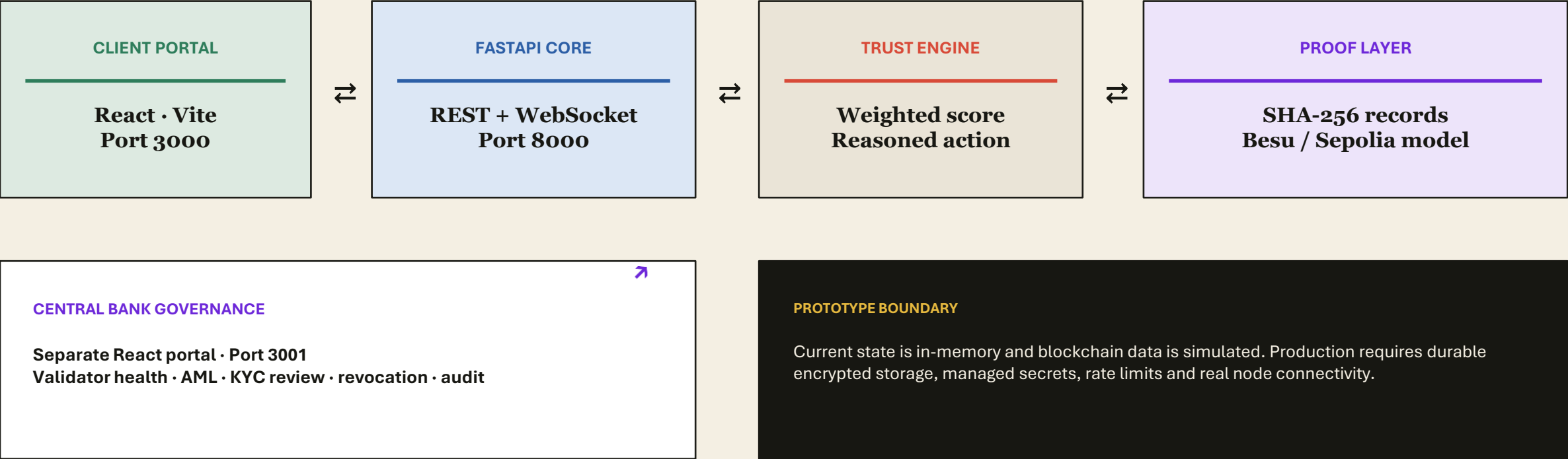
Code-backed weights: behavior 35% · device 35% · context 30%.

CLEAR THRESHOLDS MAKE THE PROTOTYPE TESTABLE TODAY—AND TUNABLE WITH REAL DATA LATER.

ARCHITECTURE

A trust layer around the bank—not a replacement for it.

REST handles banking actions. WebSockets stream live trust. Portal isolation keeps customer and regulator privileges separate.



REGULATOR VIEW

Customers bank. Regulators govern. The roles never blur.

A separate command center gives authorized teams visibility and intervention without exposing controls to customers.

VALIDATORS

Node health + consensus

AML + KYC

Alerts + review queue

REVOCATION

Freeze credentials

AUDIT

Trace every decision

Regulatory Node

RESERVE BANK GOVERNOR · PORT 3001

COMMAND CENTER

Regulatory Dashboard

BLOCKCHAIN INFRASTRUCTURE

Validator Nodes

Smart Contracts IDE

SECURITY & AML

AML & Sanction Screening

PLATFORM MANAGEMENT

Customer Assets Directory

Identity Review Queue

Live Banking Sessions

Regulatory Audit Ledger

Regulatory Console

PBFT Consensus → Live Governance

12 validator nodes →

Nexus Regulatory & Compliance Console

Reserve Bank Governor Node · Hyperledger Besu PBFT · Port 3001

ADMIN PORTAL ONLY

Live Sync: 10:41:42 AM

Central Bank Governance Portal: Signed in as Regulatory Node Governor (regulator). Regulatory authority to govern Hyperledger Besu nodes & smart contracts.

RESERVE BANK REGULATORY NODE COMMAND CENTER

Hyperledger Besu Enterprise Consortium

Consensus: IBFT 2.0 PBFT (Instant Finality) Block Height: #1489205

NETWORK: 100% OPERATIONAL

NETWORK THROUGHPUT

3450 TPS

Sub-second PBFT Block Latency

ACTIVE VALIDATOR NODES

4 Nodes

RBI, HDFC, SBI, ICICI Synced

ZK PROVER ENGINE

EIP-712 Plonk ZK-SNARK

EIP-712 Signature Enforcement

Enterprise Validator Node Topology

PBFT Consensus

VAL-RBI-NODE-01

Reserve Bank of India Node

Latency: 4ms

ONLINE

VAL-HDFC-NODE-02

HDFC Enterprise Node

Latency: 7ms

ONLINE

VAL-SBI-NODE-03

State Bank of India Validator

Latency: 6ms

ONLINE

VAL-ICICI-NODE-04

ICICI Blockchain Gateway

Latency: 8ms

ONLINE

REAL SCREENSHOT / GOVERNANCE PORTAL / PORT 3001

INNOVATION

Not another banking app. A missing trust layer.

The advantage is not one feature—it is the way identity, action and evidence stay connected.

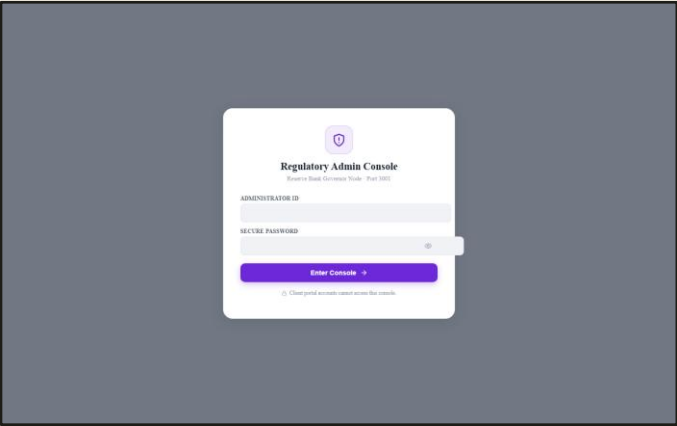
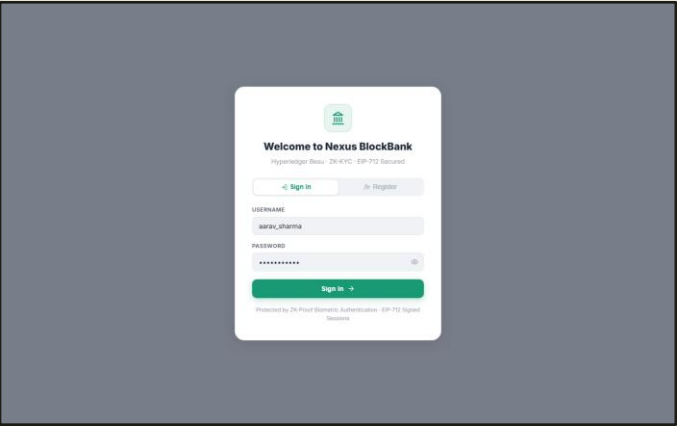
COMPARISON	CONVENTIONAL DIGITAL BANKING	NEXUS BLOCKBANK
WHEN TRUST IS CHECKED	At login	Continuously during the session
HOW RISK IS HANDLED	Generic block or alert	Allow · step-up · restrict · revoke
WHAT THE TEAM SEES	A score without context	Reason codes + action history
WHO CONTROLS WHAT	Shared operational surface	Isolated customer and regulator portals
WHAT GETS RECORDED	Mutable application log	Hash-backed proof record in the prototype

DIFFERENTIATOR: TRUST FOLLOWS THE ACTION—AND THE EVIDENCE FOLLOWS THE TRUST.

DEMO EVIDENCE

Two live portals. One tested backend.

The repository runs as a public hackathon demonstration with real screens, authenticated flows and role isolation.



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AUTOMATED SUITES PASSED

Authenticated customer flow
Admin / customer isolation
Anonymous-access rejection
Transaction input validation

<15 ms

TRUST EVALUATION

84 ms

P95 FINALITY

3,450

TPS MODEL

4

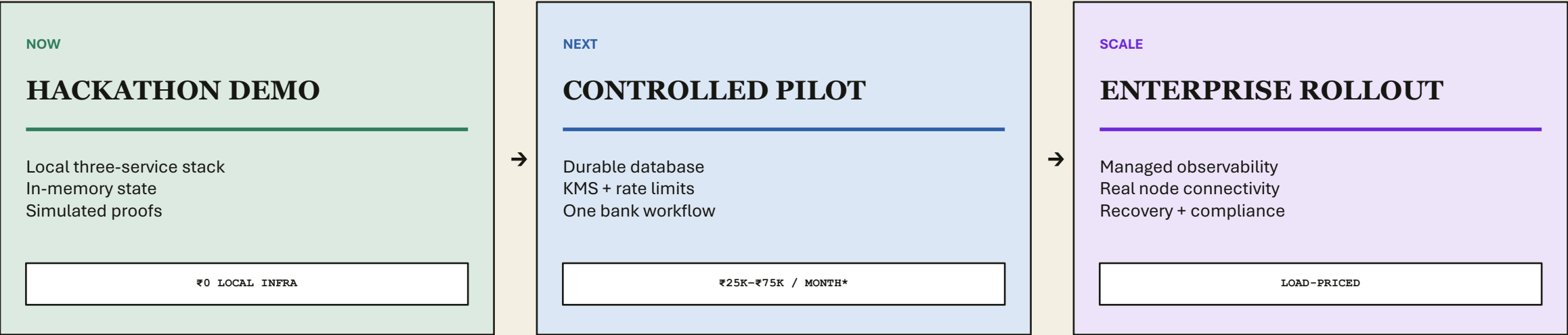
VALIDATOR NODES

HONEST LABEL: PERFORMANCE, THROUGHPUT AND BLOCKCHAIN FIGURES ARE REPOSITORY DEMO TARGETS—NOT PRODUCTION BENCHMARKS.

FEASIBILITY

Start beside the bank. Prove value. Then integrate.

The lowest-risk rollout begins as a decision layer for selected sensitive actions—not a core-banking replacement.

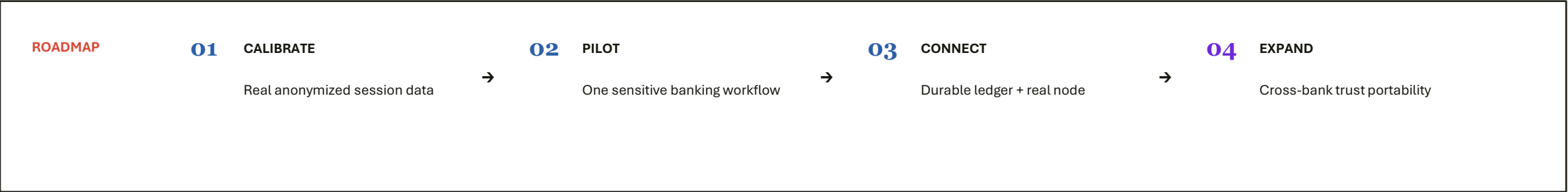
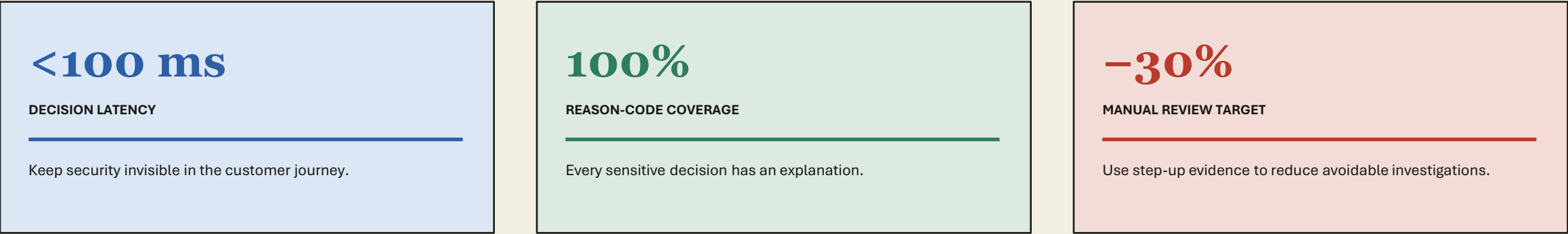


DEPLOYMENT GATES	01 Encrypted durable storage	02 Signed wallet nonce	03 Rate limits + lockout	04 Load / recovery / security tests
*Indicative small-pilot cloud estimate; validate after load testing and vendor selection.				

IMPACT

Safer actions. Fewer false alarms. Faster answers.

The pilot should be judged by measurable outcomes—not by blockchain vocabulary.



PROPOSED PILOT KPIS—TO BE VALIDATED WITH REAL BANK TRAFFIC.

THE CLOSING ARGUMENT

Trust every action.
Not just every login.

Nexus BlockBank makes digital banking safer without making it feel harder.

DEMO READY

Client 3000 · API 8000 · Admin 3001

Prashant / Project Lead & Core Developer

LIVE DECISION PATH

